

**FRESH BOWL HORTICULTURE PVT LTD**

**QUALITY ANALYSIS TESTS**

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## TESTS:

### 1. Moisture:

**Principle:** Determination of moisture content present in a material by measuring the weight loss after drying by using the indirect method.

**Apparatus:** Empty paper, Beaker, Desiccator, weighing balance and Micro oven.

**Procedure:**

- Take 100g of sample in 250 ml beaker.
- Keep it in micro oven at 72°C for maximum 12-15 min.
- Take out the sample and weigh the sample without beaker weight.

**Spawn procedure:**

- Take 50g of spawn sample in 250 ml beaker
- Keep it in hot air oven at 100°C for 2 hours
- Take out the sample and weigh the sample without beaker weight.

**Calculation:** Moisture = [(Initial weight – dry weight)]

### 2. pH (potential of hydrogen):

**Principle:** pH based on the measurement of hydrogen ion concentration in a solution. It indicates how acidic or basic (alkaline) a substrate is.

**Apparatus:** pH meter, Beaker, Glass rod

**Procedure:**

- Take 10 grams of sample in a 100 ml of beaker.
- Add 100 ml of distilled water and mix it well with glass and allow the PH meter in it.
- Immerse the probe into the sample and note down the reading.

**Note:** Ever day calibration has to be done for pH for accuracy.

**Calibration procedure:**

- Set the temperature in pH meter as per the weather temperature.
- 7.0 pH, in mV mode it is used to set zero adjustment.
- Take 100 ml of distilled water in 100 ml of beaker.
- Add one 7.0 pH tablet and melt it completely.
- Immerse the probe into the 7.0 buffer solution. In pH mode it sets digital read out to the value of standard buffer.

**3. Ash:**

**Principle:** Determining the inorganic residue remaining after complete oxidation of organic matter in a sample.

**Apparatus:** Hot plate, crucible, muffle furnace, desiccator, weighing balance.

**Procedure:**

- Take 5g of sample accurately in crucible.
- After that transfer it into muffle furnace at 600°C for 4 hours.
- After that take it out and cool in desiccator for 30 minutes.
- Then take the weight of the sample.

**Calculation:**

$$\frac{W3 - W1}{W2} \times 100$$

W1 = Crucible weight

W2 = Sample weight

W3 = Crucible + Ash weight

## 4. Nitrogen:

**Principle:** It involves converting organic nitrogen into ammonium sulphate through digestion with concentrated sulfuric acid, followed by distillation to release ammonia, which is then titrated with a standard acid to determine the nitrogen concentration.

➤ The method developed for nitrogen analysis consists of three phases:

1. Digestion
2. Distillation
3. Titration

**Apparatus:** Nitrogen Estimation system (Block digestion system with distillation system), Digestion tube, conical flask, burette, burette stand, pipette, 10 ml of test tube.

### Reagents:

1. Sodium hydroxide solution
2. Boric acid extra pure
3. Mixed indicator (ethanol-400ml, methyl red-2gms, bromocresol green- 0.3gms).
4. Hydrochloric acid
5. Sulphuric acid (98%)
6. Catalyst (Potassium sulphate-500gms, cupric sulphate- 25gms)

### Reagents preparation:

#### 1. Sodium hydroxide solution:

- Take 500ml of distilled water in a 1000 ml of beaker.
- Add 200 grams of Sodium hydroxide pellets in it and mix it well with glass rod.

#### 2. Boric acid solution:

- Take 500ml of distilled water in a 1000 ml of beaker.
- Add 20 grams of boric acid powder in it and mix it well with glass rod and add 10ml of mixed indicator in it.

### **3. Titrant solution:**

- Take 500ml of distilled water in a 1000 ml of beaker.
- Add 4.3 ml of Hydrochloric acid with 10 ml of pipette.

## **Procedure:**

### **1. Digestion:**

Guidelines to operate Block Digestion system.

**STEP 1:** Switch on the Digestion system and set the temperature controller at 100°C

**STEP 2:** Once the initial temperature of 100°C is reached reset the temperature controller at the recommended digestion temperature

**Recommended Digestion Temperature:** 350°C - 420°C

**STEP 3:** Add 3grams of Catalyst mixture after thoroughly Sterilizing the digestion tube. Add the 200 grams of sample and 10 ml of Sulphuric acid. Use the special stainless steel insert rack for handling digestion tubes.

**STEP 4:** Place the SS Insert Rack with sample on the insert rack stand. Fix the exhaust manifold system on the top of the insert rack.

**STEP 5:** Open the water line for the suction system.

**STEP 6:** Once the set Digestion temperature is reached, load the insert rack with the samples and exhaust manifold system into the digestion block.

**STEP 7:** Adjust the water flow to the suction system, so that it generates sufficient vacuum for absolute removal of acid fumes.

**STEP 8:** During initial stages of digestion dense acid fumes will evolve, which will be efficiently trapped by the exhaust system. After sometime the fumes reduce and the condensation starts on the walls of the digestion tubes.

At this stage reduce the water flow to suction system (switch off the monoblock pump when pump is used) so that minimum suction exists. This avoids evaporation of acid and enables efficient digestion.

**STEP 9:** After the digestion is completed, lift the insert rack with the digested samples and exhaust manifold system and place it on the insert rack stand. In this cooling phase, samples get cooled. Leave the suction system with minimum water flow. This helps in fast cooling and also removes scarce acid fumes.

**STEP 10:** When the samples are cooled, put off the water flow in the suction system.

**STEP 11:** Take the digested samples in the insert rack for distillation

#### **DO'S:**

1. Always do handle the digestion tubes with the specially designed SS Insert rack only. This avoids breakage of digestion tubes.
2. Do ensure sufficient water level in the overhead tank throughout the digestion.
3. Do drain out the condensed acid collected in the suction pumps periodically.
4. Do use clear water without any dust particles for the suction pump system. Dust/mud in water may block the flow in suction pump and lead to serious damages.
5. If the monoblock pump is installed with the suction system do switch on the monoblock pump every day and run it for at least 10 minutes even if the digestion system is not regularly used. This will prevent the motor getting struck and keep the pump fully functional.

#### **DONT'S:**

1. Do not exceed recommended digestion temperature limits during digestion. Carrying out digestion with temperature higher than the recommended limits is unsafe, and may cause explosion especially with Di acid (or) Tri acid application.
2. DO not attempt to remove manifold system during digestion. In the effort you may break the exhaust manifolds.
3. Do not use the digestion system with unstablised electric power. At low voltage the system may take long time for heating up and higher voltage may damage the heaters.

## **2. Distillation**

### **Step: 1**

- Switch 'ON' the Main power at the back of the system.
- Switch on the power switch in the control panel.
- Keep the receiver solution on the receiver end.
- Load the sample in to the distillation system by operating the side lever.

### **Step: 2**

- Keep the, 500 ml of Boric Acid and 500 ml of sodium hydroxide, reservoirs to the rear of the system. Reagents can also be added manually.
- Immerse the hose coming out from the system in to the respective reservoir tanks.

### **Step: 3**

- Wait for the "READY" indication in the Control Panel. Select the process time in the control panel. Open the water inlet to the water condenser.
- Switch 'ON' the "Process Switch"
- Open the water inlet to the water condenser.
- Ensure Addition of steam in the tube.

### **Step: 4**

- Take out the sample one by one and dilute with 20 ml of distilled water.
- Dilution is necessary to avoid solidification of sample on cooling.
- The diluted sample is placed in the distillation system for Ammonia Recovery
- Water Refilling to the boiler takes place automatically. During refilling of water ensure an alarm and an indication. Ensure the distilled water reservoir always filled with sufficient distilled water.
- Select the program number one by operating the PGM Key.

- Press RUN' key to activate the program. Ensure addition of Reagents and Steam. Ensure collection of ammonia in the receiver solution.

### 3. Titration:

- Remove the receiver solution with ammonia and take to a manual burette for titration.
- Titration with titrant solution to estimate the Nitrogen Percentage.

#### Calculation:

$$\text{Nitrogen} = \frac{14.01 \times 0.1 \times (\text{Titrate value} - \text{Blank value}) \times 100}{\text{Sample weight} \times 100}$$

### 5. Carbon Nitrogen Ratio:

**Principle:** The C.N ratio is the mass of carbon to the mass of nitrogen in a particular substance.

#### Calculation:

$$\text{CN ratio} = 100 - \text{Ash} \times 0.5/\text{N}$$

## 6. EC (Electro Conductivity):

**Principle:** Conductivity is a measure of the ability of an aqueous solution to carry an electric current.

**Apparatus:** EC meter, Beaker, Glass rod

### **Procedure:**

- Take 10 grams of sample in a 100 ml of beaker.
- Add 100 ml of distilled water and mix it well with glass.
- Immerse the bottom of the tester into the sample and note down the reading.

**Note:** Weekly once calibration has to be done for EC for accuracy.

## 7. TDS (Total dissolved solids):

**Principle:** The total amount of **minerals salts** and other substances dissolved in liquid.

- Shows the electrical conductance of a solution.
- Measured in terms of Micro Siemen or ppm.

### **Calculation:**

$$\text{TDS} = \text{EC} \times 0.64$$

## 8. Total hardness:

**Principle:** Total hardness test determines the **Chlorides, Sulphates and Nitrates of Calcium and Magnesium** present in water.

**Apparatus:** Total Hardness kit.

**Procedure:** Follow as per Total Hardness kit procedure.

| S.no | Raw Material      | Moisture | pH      | Ash    | Nitrogen | Carbon & Nitrogen ratio |
|------|-------------------|----------|---------|--------|----------|-------------------------|
| 1    | Paddy straw       | 12-13%   | 7.0-7.3 | 12-17% | 0.7-1.0  | 60-70                   |
| 2    | Mustard straw     | 7.5-8%   |         | 6 - 8% | 0.5-0.9  | 51-60                   |
| 3    | Bagasse           | 40-55%   | 4.6-5.0 | 2-5%   | 0.4-0.5  | 97-98                   |
| 4    | Chicken Manure    | 16-20%   | 7.5-8.0 | 25-33  | 2.7-3.5  | 11-15                   |
| 5    | Gypsum            | 25-30    | 2.8-3.5 | 85-88  | -        | -                       |
| 6    | Urea              | -        | -       | -      | 46       | -                       |
| 7    | Ammonium sulphate | -        | -       | -      | 20.6     | -                       |

**Table: 1 Quality test ranges of raw material in Compost**

| S.no | Compost procedure Phase  | Moisture  | pH      | Ash   | Nitrogen | Carbon & Nitrogen ratio |
|------|--------------------------|-----------|---------|-------|----------|-------------------------|
| 1    | Soak pit water           | -         | 6.5-7.5 | -     | -        | -                       |
| 2    | Bagasse wetting          | 75 – 78   | -       | -     | -        | -                       |
| 3    | Bagasse bunker loading   | 75 – 77   | -       | -     | -        | -                       |
| 4    | Bagasse reloading        | 75 – 76.5 | -       | -     | -        | -                       |
| 5    | Turning/0+Turning/1      | 72 – 74   | -       | -     | -        | -                       |
| 6    | Turning/2+Bunker loading | 74 – 77   | 8.1-8.4 | 16-20 | 1.3-1.6  | 25-32                   |
| 7    | Reloading                | 75 – 76.5 | 7.8-8.0 | 19-23 | 1.4-1.7  | 22-28                   |
| 8    | Tunnel loading           | 72.5 – 74 | 7.6-7.8 | 23-26 | 1.6-1.9  | 19-24                   |
| 9    | Growing room loading     | 65 – 68   | 7.4-7.7 | 25-29 | 1.8-2.2  | 16-20                   |

**Table: 2 Quality test ranges of phases in Compost**

| S.no | Water               | PH        | EC        | TDS       | TH            |
|------|---------------------|-----------|-----------|-----------|---------------|
| 1    | Bore well water     | 6.0-7.30  | 700 – 900 | EC x 0.64 | 250 – 400 ppm |
| 2    | Maintenance water   | 4.5–6.0   | 110 - 150 | EC x 0.64 | 4 – 20 ppm    |
| 3    | Chilling line water | 8.0 - 9.5 | 400 – 600 | EC x 0.64 | 16 – 30 ppm   |
| 4    | Cooling tower water | 8.8 - 9.8 | 4.2 – 6.5 | EC x 0.64 | 300 – 500 ppm |
| 5    | Soft water          | 7.0 - 8.0 | 500 - 900 | EC x 0.64 | 20 – 200 ppm  |
| 6    | Boiler water        | 7.5 - 9.0 | 700 – 900 | EC x 0.64 | 10 – 100 ppm  |
| 7    | Compost RO-water    | 4.5 - 6.0 | 050 – 150 | EC x 0.64 | 10 – 50 ppm   |
| 8    | Growing RO-water    | 5.0 - 6.0 | 100 – 200 | EC x 0.64 | 4 – 50 ppm    |
| 9    | Canteen RO-water    | 5.0 – 6.5 | 050 - 150 | EC x 0.64 | 10 – 50 ppm   |

**Table: 3 Quality test ranges of Water**

| Material                  | PH      | Moisture | EC        |
|---------------------------|---------|----------|-----------|
| Casing soil( mixing)      | 7.8-8.1 | 67 – 70  | 200 – 300 |
| Casing soil( application) | 7.6-7.8 | 65- 68   | 170 -250  |

**Table: 4 Quality test ranges of Casing soil**

| Material | Moisture | PH      |
|----------|----------|---------|
| Spawn    | 40-45    | 7.3-7.7 |

**Table: 5 Quality test ranges of Spawn**